

FurFeel Flutter Mobile QA Findings

Simple summary of the latest QA pass on auth boundaries, live telemetry, owner copy, and core Home flows.

Quick Summary

The most important problem found is a layout overflow in the floating bottom navigation. It happens when the app reaches the main owner shell and can break widget tests before Home finishes rendering. The auth redirect boundary checks looked good in the focused tests: Google OAuth is pinned away from the dashboard domain.

Highest priority	Fix the floating navigation overflow.
Why it matters	A tiny visual overflow can become clipped text, broken accessibility, or unstable tests on smaller screens.
Second item	Clarify the new Updated time so users know whether it refers to vitals telemetry or the stress score.
Production data	No production data was touched during this QA pass.

What Was Checked

- Google OAuth redirect logic, including checks that mobile web does not return to furfeel.site.
- Mobile access role guard, including admin/staff denial in the owner app.
- No-diagnosis copy checks for owner-facing language.
- Consent gate flow into the main app shell.
- Live telemetry timestamp display, including the new AM/PM format.
- Static analysis on the mobile app and focused tests.

Findings

Finding 1: Floating bottom nav overflows on Home shell

Severity: High

Area: Accessibility / UX

Plain explanation: The bottom navigation is just a little too tall for the space it gives itself. Flutter reports a 2 pixel overflow. That sounds small, but it can cause clipped labels, unstable widget tests, and problems on small screens or larger text settings.

Steps	Run <code>flutter test test/consent_gate_test.dart</code> and let the app unlock into RootShell.
Expected	The app lands on Home and shows Health overview with no layout exceptions.
Actual	Flutter throws RenderFlex overflowed by 2.0 pixels on the bottom . The test then cannot find Health overview.
Evidence	Overflow points to <code>apps/mobile/lib/widgets/floating_nav_bar.dart:192</code> . The nav surface is fixed at 64 px high at line 112, while the item contains vertical padding, icon, spacing, and a text label.
Device	Flutter widget test environment on local macOS.
Restart	Reproduces from a fresh test launch. Hot restart/full restart on a physical device was not tested.

Recommended next action: Reduce the vertical content inside each nav item, increase the nav surface height, or make the item layout flexible. Then rerun consent, root shell, and accessibility text-scale tests.

Finding 2: Updated time may be unclear when telemetry and classification arrive separately

Severity: Medium

Area: Home / Telemetry

Plain explanation: The Home hero now shows a helpful time like Updated 3:42 PM. The issue is wording: the time comes from the newest telemetry reading, but the stress score comes from a separate classification event. For a short moment, the time can update before the stress score updates.

Steps	Receive a realtime telemetry_readings insert before the matching stress_classifications insert.
Expected	The UI makes clear whether the time refers to vitals telemetry or the stress score.
Actual	The hero can show the previous stress level while the chip says Updated 3:42 PM from the newer reading.
Evidence	Home passes <code>widget.reading?.capturedAt</code> into the hero at <code>apps/mobile/lib/screens/home/home_tab.dart:115</code> . Realtime subscribes to telemetry_readings and stress_classifications separately in <code>furfeel_repository.dart:880-884</code> .
Device	Static/code QA based on realtime event ordering.
Restart	Not restart-dependent. It depends on event ordering.

Recommended next action: Rename the chip to something more specific, such as Last vitals 3:42 PM, or show separate timestamps for vitals and stress score.

Passing Checks

- Google OAuth redirect tests passed, including dashboard-domain denial.
- Mobile access role test passed: only owner accounts are allowed into the mobile app.
- No-diagnosis copy test passed.
- Login page tests passed in the focused batch.
- AM/PM timestamp formatter test passed.
- Flutter analyze on the checked mobile files passed.

Suggested Fix Order

- First: fix the floating navigation overflow, because it blocks reliable Home/consent QA and can affect small screens.
- Second: clarify the Updated chip wording so owners do not confuse telemetry freshness with stress-score freshness.
- Third: rerun `root_shell_test.dart`, `consent_gate_test.dart`, a11y text scale tests, Google OAuth tests, and no-diagnosis copy tests.